

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

SKILLS

Languages: English (C1 [IELTS 8.0](#) )
Programming: Python, SQL, Bash, Julia, C/C++, R
Tools: Linux, Git, Docker, pandas, PyTorch, sklearn, opencv, open3d, Quarto, L^AT_EX

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: [Márton Vaitkus](#) 
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 (**rank 3/34**)
Supervisor: [Galtbayar Artbazar](#) 
Thesis: [Evaluating land degradation using image processing methods](#) 

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- **Lab Lead:** Authored and maintained comprehensive lab documentation and user guides, enabling onboarding for new students and establishing clear protocols for research operations. Served as Lab Lead for a team of 6-8 students; collaborated to present complex mathematical concepts during weekly seminars.
- **Data Management:** Architected a unified, well-documented data folder structure for research backups, directly improving data retrieval efficiency and creating written guidelines for future data integration.
- **Research:** Participated in a field survey drone imagery captures. Developed a custom [pipeline](#)  for processing drone imagery which is used to obtain the necessary vegetation index from drone imagery data. Worked in the Linux Environment with version control to ensure reproducibility.
- **Contributions:** Incorporated an open-source Docker containerized application (OpenDroneMap) for drone ortho-mosaic processing instead of proprietary software.

PROJECTS

[aiSim Data Integration for 3D Gaussian Splatting](#)

- Developed a sensor fusion pipeline to parse and convert over sensor measurements from aiMotive's [aiSim](#)  ADAS simulation platform into the [nerfstudio](#)  3D scene representation format. Worked on the camera coordinate system transformation from sensor frame to the world frame.

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced polynomial regression experiments from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks to using interactive sliders.

[Exploratory Data Analysis – Bike Sharing Dataset](#)

- Performed exploratory data analysis on bike sharing dataset. Published the results on the personal blog using Quarto.

AWARDS

Stipendium Hungaricum Scholarship Sep, 2025